Algebraic Notation

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Algebraic notation

What is algebra?

- **Algebra** is a topic in mathematics that uses **letters** to represent **general** (or **unknown**) **numbers**
  - x and y are two unknown numbers
    - More information is needed to find their values
- Letters are also called **variables**

How do I write calculations in algebra?

- Writing mathematical ideas in letters is called using **algebraic notation**
- You can use + and – to mean **add** and **subtract**
  - $a + b$
    $c + d - e$
- However **no** symbol is used for **multiplication**
  - ab (means $a \times b$)
    $3ab$ (means $3 \times a \times b$)
- **Fractions** are used for **division**
  - $\frac{a}{b}$ (means $a \div b$)
- You can combine the ideas above
  - $ab + \frac{c}{3}$ (means $a \times b + c \div 3$)
  - The **order of operations** for numbers still works
    - work out $a \times b$ and $c \div 3$ before adding them together
- **Powers** (**indices**) and **roots** are the same as with numbers
  - a^2 means $a \times a$
    $4a^2$ means $4 \times a^2$
    - With the order of operations, a^2 happens before multiplying by 4
  - $\sqrt{a}$ means the square root of a
- **Brackets** work in the same way as they do with numbers
  - $3(a + b)$ means $3 \times (a + b)$
    - Add the a and b first, then multiply the result by 3

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

Worked Example

Raheem is playing a game and starts with q points.
He then earns a further 6 points.

(a) Use algebra to write down the total number of points that Raheem has.

Answer:

Raheem has $q + 6$ points

The game then decides to doubles his total number of points.

(b) Use algebra to write down the number of points that Raheem now has.

Answer:

Raheem has $2(q + 6)$ points

You can also write:

Raheem has $(2q + 12)$ points

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